**SIMATS ENGINEERING**
**Saveetha Institute of Medical and Technical Sciences**
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Engineer to Excel

## ITA0328 - Mobile Computing

# Fundamentals of Mobile Computing

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# INTRODUCTION

- Mobile computing is a technology that allows users to access information and communicate while moving from one place to another.
- It uses mobile devices such as smartphones, tablets and laptops.
- It enables wireless data transfer through Wi-Fi, Bluetooth, 4G and 5G networks.
- It provides easy access to internet services, online applications and real-time communication.
- Mobile computing is widely used in education, healthcare, banking and business.

# Components and Characteristics

- Components

  Hardware: Smartphones, tablets and laptops.

  Software: Android, iOS and mobile applications.

  Network: Wi-Fi, Bluetooth, 4G and 5G.

- Characteristics

  Portability and mobility.

  Wireless connectivity.

  Real-time communication.

# Types and Generations of Networks

Types of Networks

WPAN: Bluetooth.

WLAN: Wi-Fi.

WMAN: Metropolitan wireless networks.

WWAN: Cellular networks.

Generations

1G: Analog voice.

2G: Digital calls and SMS.

3G: Mobile internet.

4G: High-speed internet.

5G: High speed and low latency.

# Applications and Advantages

Applications

- Mobile banking and digital payments.
- Online education and healthcare.
- GPS navigation and location tracking.
- Social media and video calling.

Advantages

- Easy communication.
- Quick access to information.
- Supports remote learning and working.
- Saves time and improves convenience.

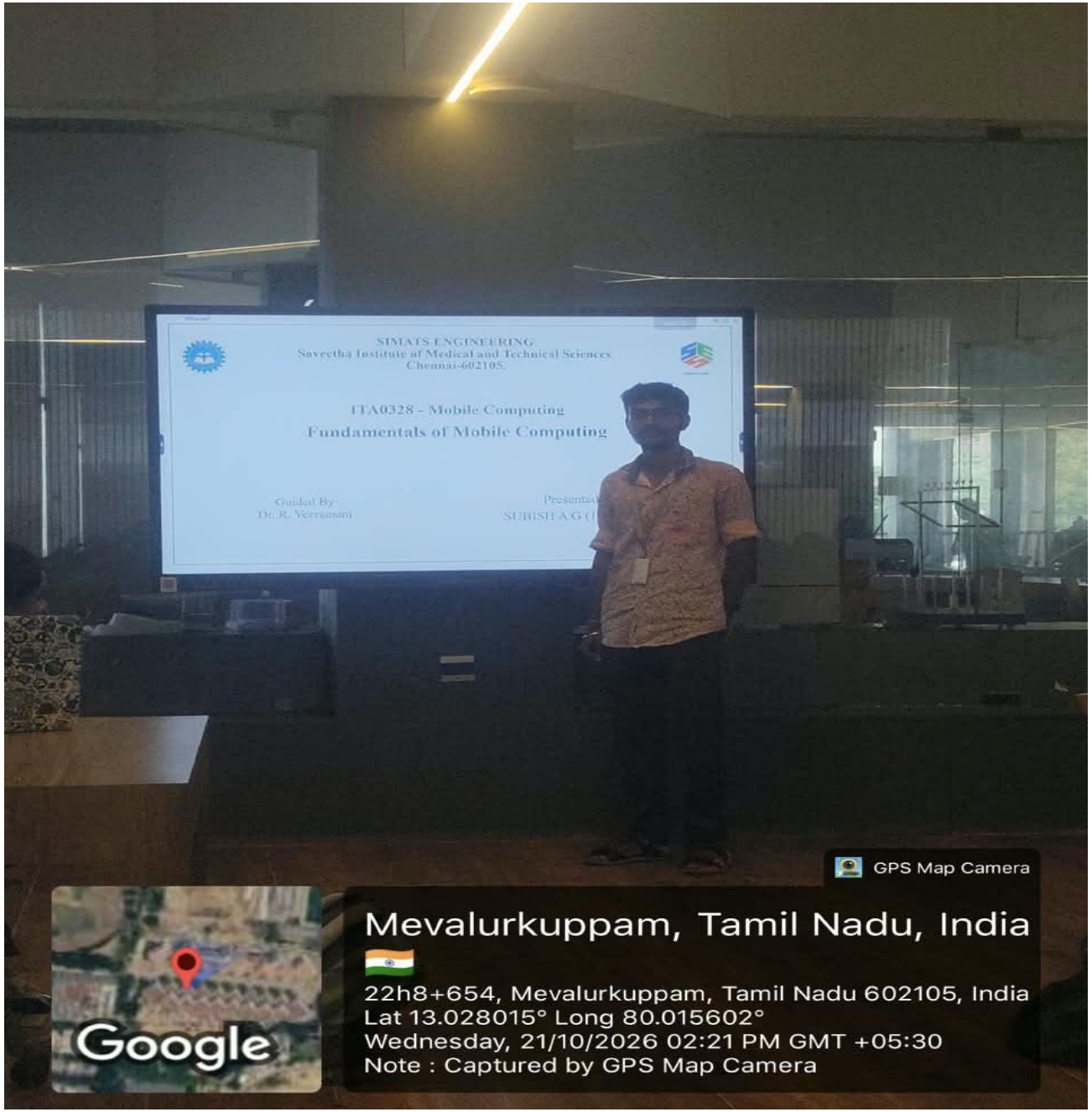
# Challenges and Conclusion

Challenges

- Security and privacy risks.
- Limited battery life.
- Network coverage problems.
- Signal interference and connectivity issues.

Conclusion

- Mobile computing is an important part of modern life.
- It combines mobile devices, software and wireless networks.
- Advancements in 4G and 5G make mobile services faster and more reliable.


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